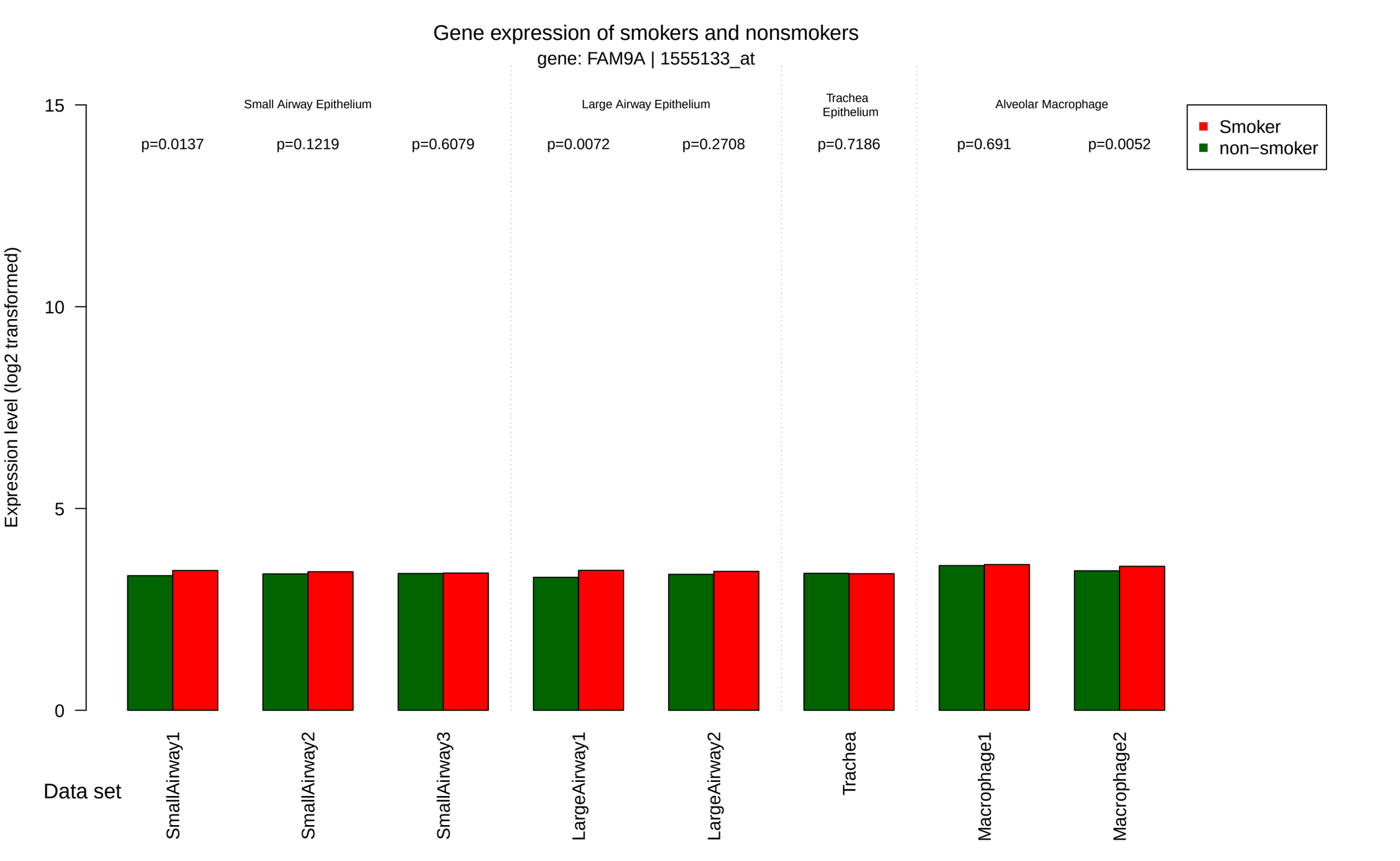
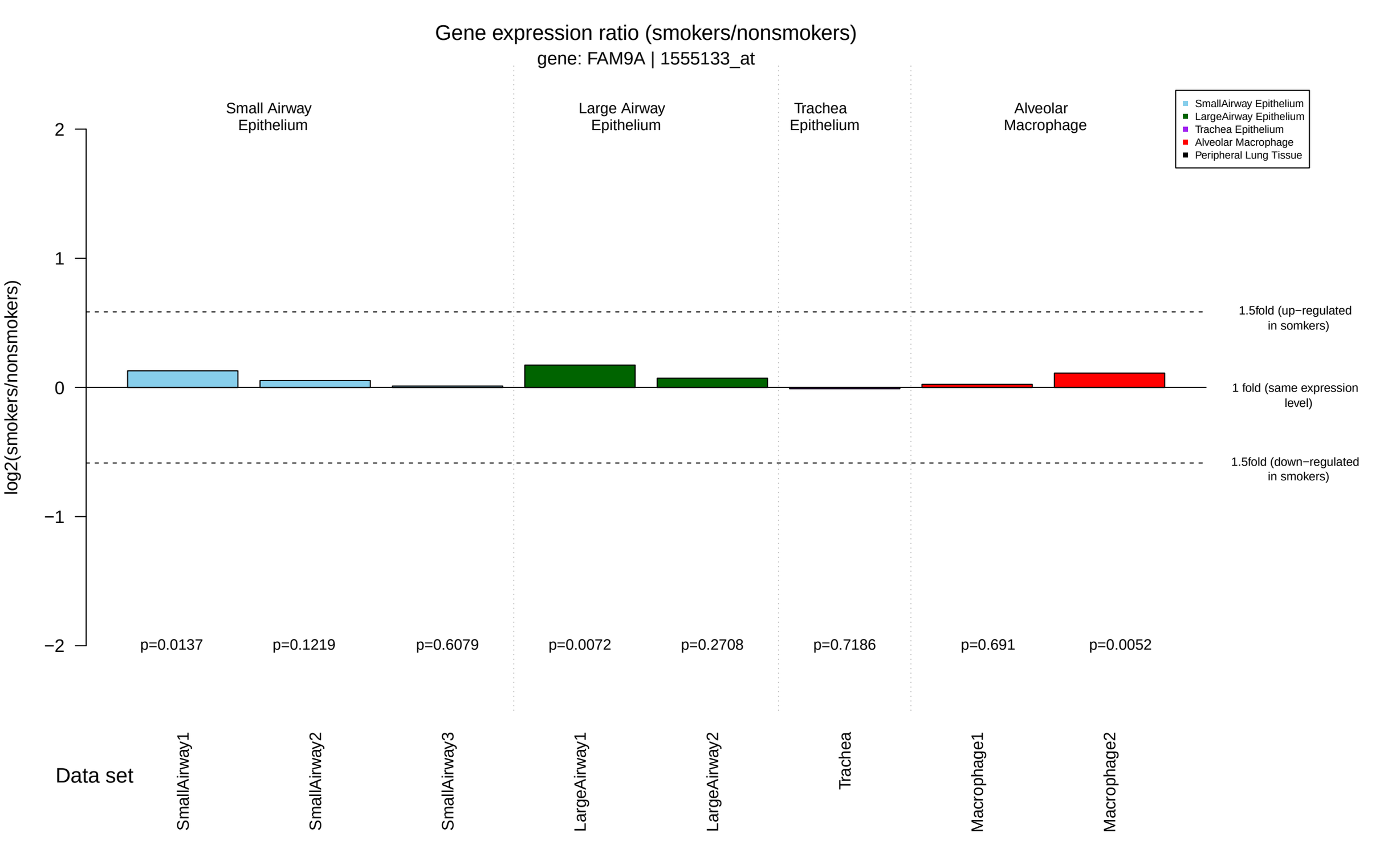
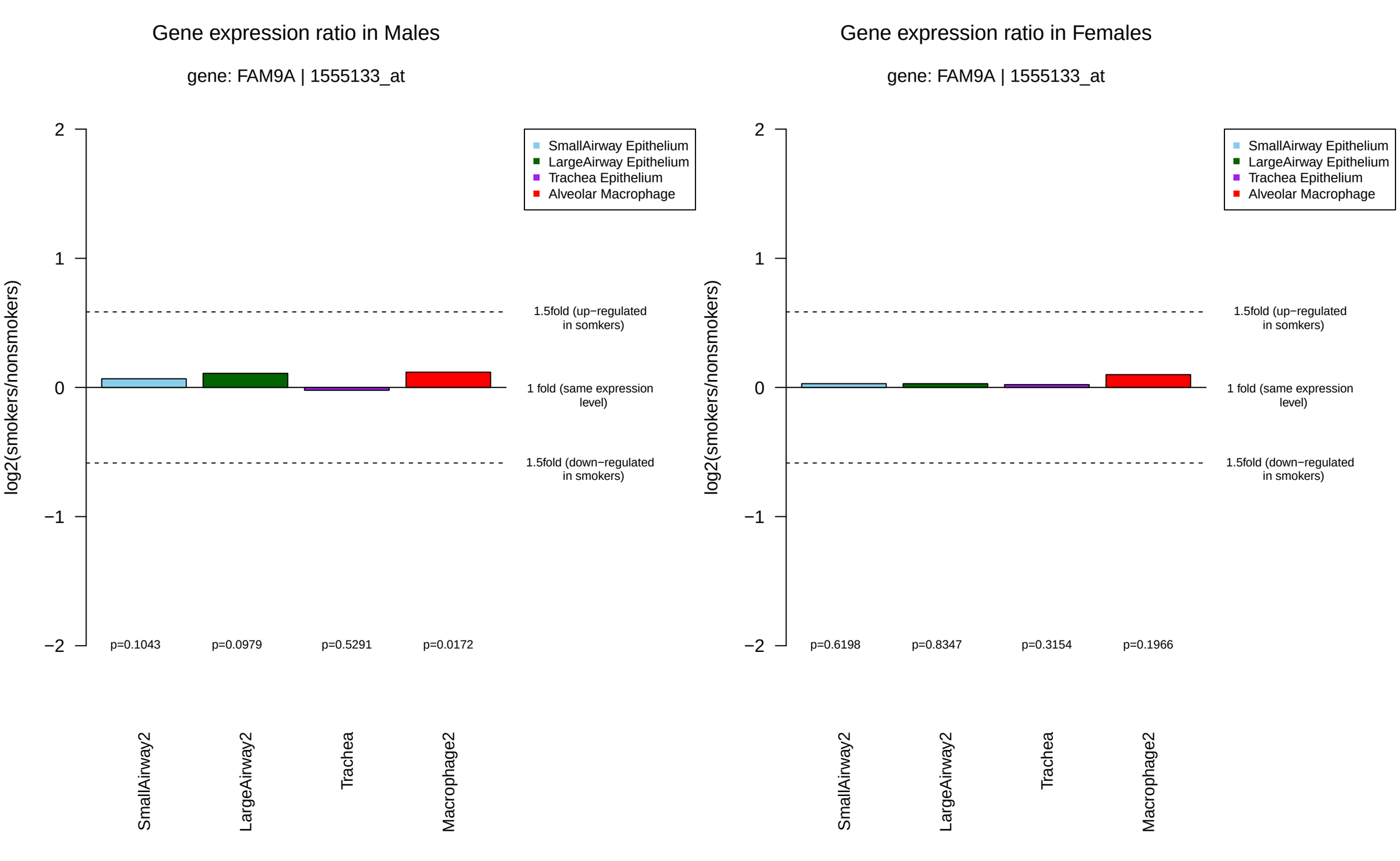
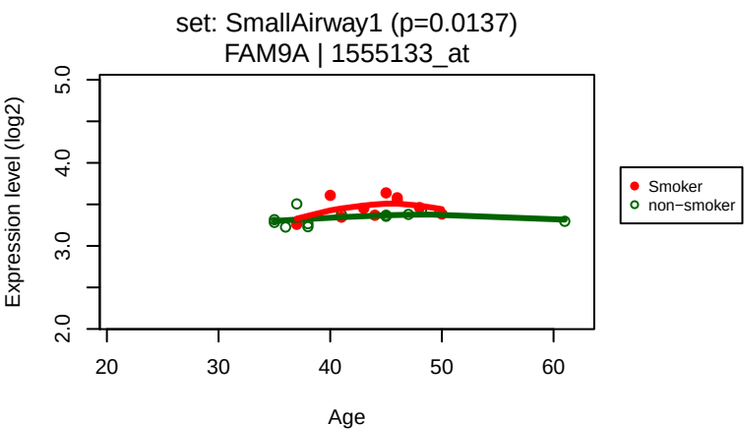
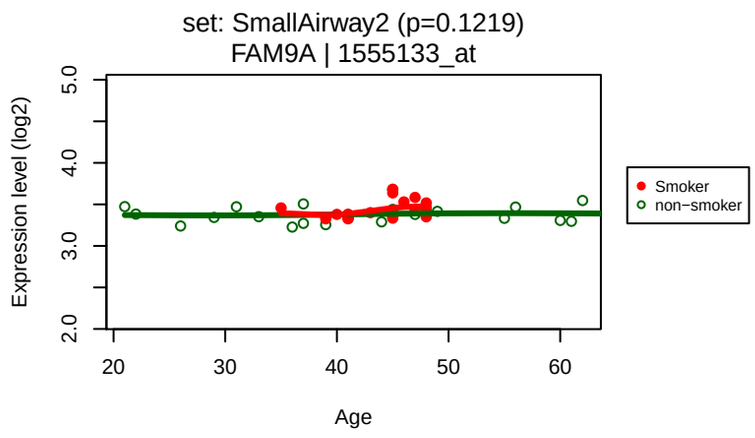
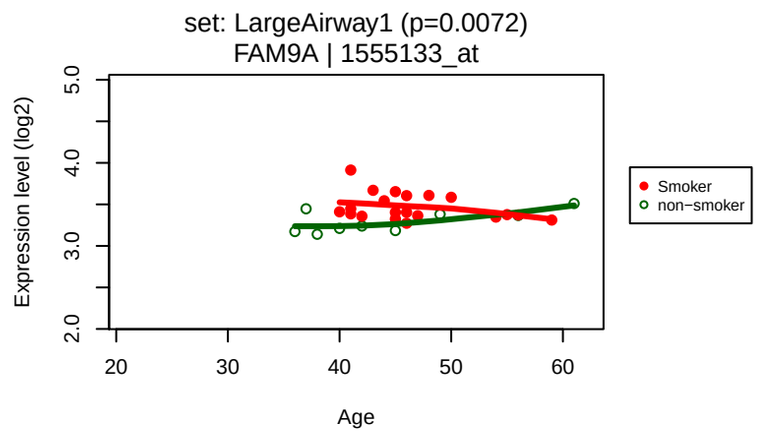

Gene expression of smokers and nonsmokers
gene: FAM9A | 1555133_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0137
p=0.1219
p=0.6079
p=0.0072
p=0.2708
p=0.7186
p=0.691
p=0.0052
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

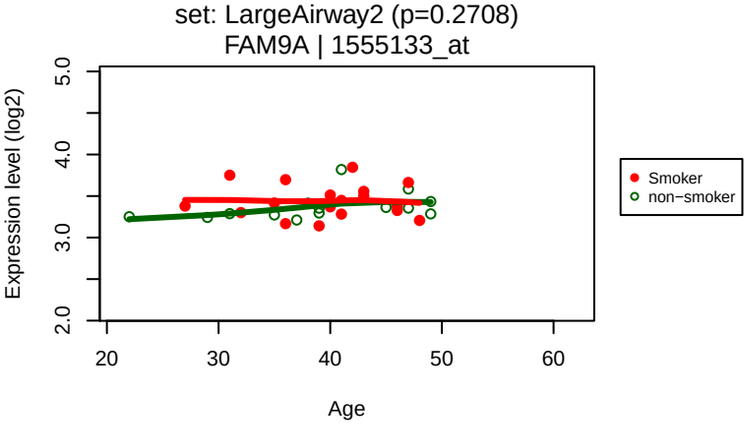

Gene expression ratio (smokers/nonsmokers)
gene: FAM9A | 1555133_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0137
p=0.1219
p=0.6079
p=0.0072
p=0.2708
p=0.7186
p=0.691
p=0.0052
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

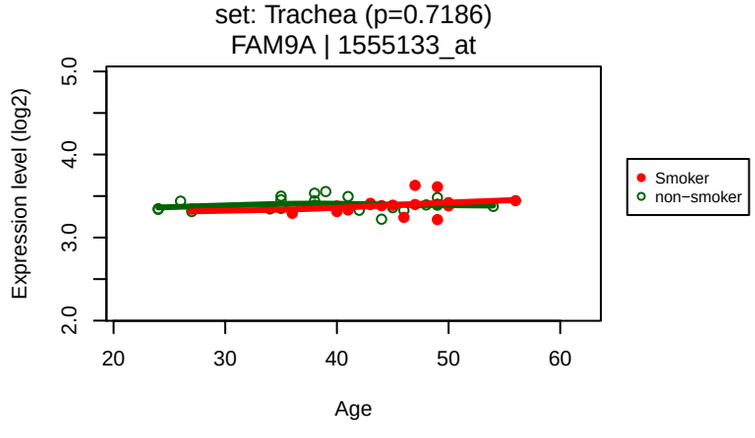

Gene expression ratio in Males
gene: FAM9A | 1555133_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1043
p=0.0979
p=0.5291
p=0.0172
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FAM9A | 1555133_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6198
p=0.8347
p=0.3154
p=0.1966
SmallAirway2
LargeAirway2
Trachea
Macrophage2

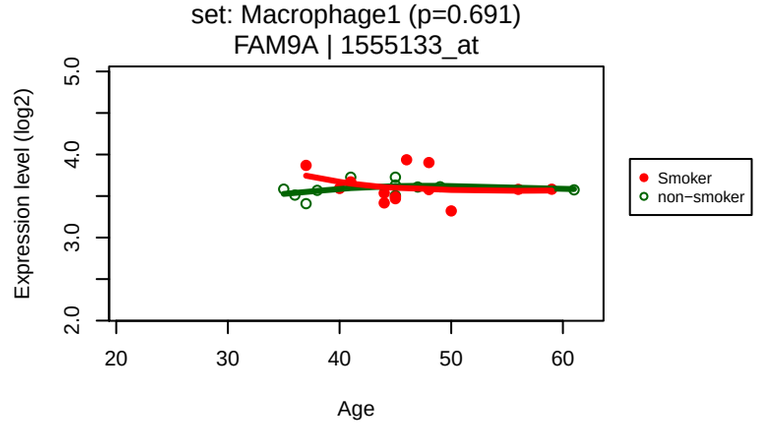

set: SmallAirway1 (p=0.0137)
FAM9A | 1555133_at
Expression level (log2)
Age
Smoker
non−smoker

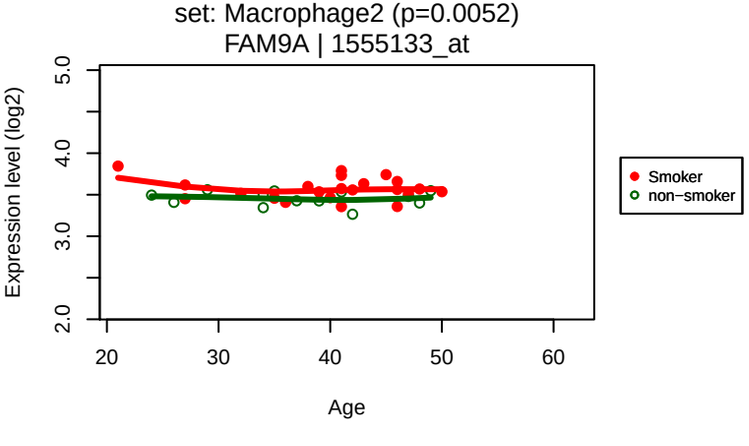

set: SmallAirway2 (p=0.1219)
FAM9A | 1555133_at
Expression level (log2)
Age
Smoker
non−smoker

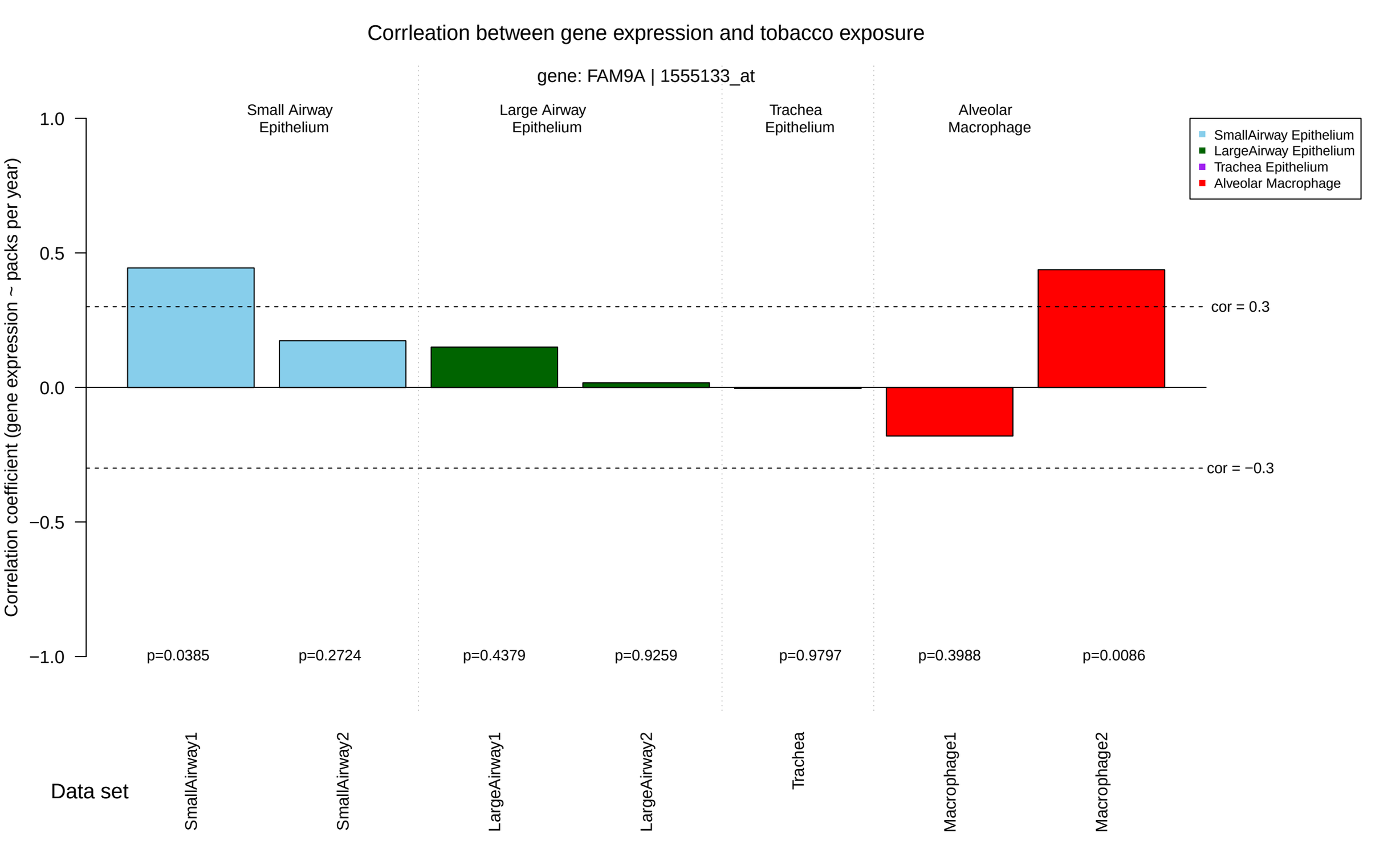

set: LargeAirway1 (p=0.0072)
FAM9A | 1555133_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2708)
FAM9A | 1555133_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7186)
FAM9A | 1555133_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.691)
FAM9A | 1555133_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0052)
FAM9A | 1555133_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FAM9A | 1555133_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0385
p=0.2724
p=0.4379
p=0.9259
p=0.9797
p=0.3988
p=0.0086
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2